पोलीस उप अधीक्षक / सहायक पोलीस आयुक्त, दळणवळण व माहिती तंत्रज्ञान विभाग, महाराष्ट्र राज्य पोलीस सेवा, गट-अ

गृह विभाग

Deputy Superintendent of Police / Assistant Commissioner of Police Communication and Information Technology Department, Maharashtra State Police Service, Group-A

Home Department

परीक्षेचे टप्पे: **१)** लेखी परीक्षा - २०० गुण **२)** मुलाखत- ५० गुण

-: परीक्षा योजना :-

विषय व सांकेतांक	माध्यम	प्रश्नसंख्या	गुण	कालावधी	दर्जा	प्रश्नपत्रिकेचे स्वरुप
विभागाशी व पदाच्या कर्तव्य व जबाबदा-यांशी संबंधित घटक (११२८)	इंग्रजी	१००	२००	एक तास	पदवी	वस्तुनिष्ठ बहुपर्यायी

अ) नकारात्मक गुणदान -

१) प्रत्येक चुकीच्या उत्तराकरीता २५% किंवा १/४ एवढे गुण एकूण गुणांमधून वजा/ कमी करण्यात येतील.
२) एखाद्या प्रश्नाची एकापेक्षा अधिक उत्तरे दिली असल्यास अथवा ज्या उमेदवाराने उत्तरपत्रिकेत पूर्ण वर्तुळ चिन्हांकित केले नसेल अशा प्रश्नाचे उत्तर चुकीचे समजण्यात येऊन त्या प्रश्नाच्या उत्तराकरीता २५% किंवा १/४ एवढे गुण एकूण गुणांमधून वजा/कमी करण्यात येतील.
३) वरीलप्रमाणे कार्यपध्दतीचा अवलंब करताना एकूण अंतिम गुणांची बेरीज अपूर्णांकात आली तरीही ती अपूर्णांकातच राहील व पुढील कार्यवाही त्याच्या आधारे करण्यात येईल.
४) एखाद्या प्रश्नाचे उत्तर अनुत्तरित असेल तर, अशा प्रकरणी नकारात्मक गुणांची पध्दत लागू असणार नाही.

ब) अंतिम गुणवत्ता यादी ही वस्तुनिष्ठ परीक्षेतील व मुलाखतीतील एकत्रित गुणांवर आधारीत राहील.

-: अभ्यासक्रम :-

विभागाशी व पदाच्या कर्तव्य व जबाबदा-यांशी संबंधित घटक या विषयामध्ये खालील घटक व उपघटकांचा समावेश असेल.

Sr. No.	Topic
	Electronics Engineering
1	**Analog Electronics :** • **Introduction to Electronic Engineering** • **Ohm's Law, Kirchhoff's Laws** • **Series and Parallel Circuits** • **Basics of Semiconductors** Basics and Characteristics of diode/ Transistor Diodes for different uses: Junction and Field Effect Transistors (BJT, JFET and MOSFET) Diode circuits clipping, clamping Transistors at low and high frequencies Amplifiers, Single and multistage, Feedback amplifiers, Operational Amplifiers (Inverting, Non-inverting, summing), Instrumentation amplifier, Precision rectifier V-to-I and I-to-V converter Op-Amp based active filters Oscillators and signal generators, Function Generators and wave- shaping circuits, 555Timers, Power Supplies.
2	**Digital Circuits :** • Boolean algebra • Logic gates; Basics of Integrated Circuits (ICs), digital IC families (DTL, TTL, ECL, MOS,

	CMOS) Adders, multiplexers, de-multiplexers, decoders, PROMs and PLAs. Sequential circuits: latches and flip-flops, counters and shift-registers ADCs, DACs, Micro-processor(8085) and interfacing with peripherals 8255 and 8259 and Microcontrollers: 8051
3	**Electronic Measurements and Instrumentation** : • Principles of Measurement, accuracy, precision and standards; • Analog and Digital systems for measurement, measuring instruments for different application; • Static/ dynamic characteristics of measurements systems, errors, statistical analysis and curve fitting; • Measurement systems for non-electrical quantities; Basics of telemetry; • Different types of transducers and displays; Data acquisition system basics.
4	**Electromagnetics :** • Elements of Vector calculus : divergence and curl; • Gauss' and Stokes' theorems, Maxwell's equations: differential and integral forms. Wave equation, poynting vector. Plane waves : Propagation through various media; reflection and refraction; • Phase and group velocity; • Skin depth, Transmission lines; • Characteristics impedance; impedance transformation; smith chart; impedance matching; S-parameters, pulse excitation. Basics of propagation in dielectric waveguide and optical fibers. Basics of Antennas: Dipole antennas; radiation pattern; antenna gain.
5	**Integrated Electronics :** • Analog and mixed Signals • RF/ Microwave/ millimeter –wave circuits and systems • Components and packaging • Wireline Circuits and systems • Biosensors and Bioelectronics • Sensor Interface Circuits • Applied Electromagnetics • Circuit / Component Modeling
6	**Advanced Electronics Topics:** • VLSI technology, Processing, lithography, interconnects, packaging, testing, VLSI design, Principles, MUX/ROM/PLA – based design, Moore & Mealy circuit design, Pipeline concepts & functions; • Design for testability, examples, DSP, Discrete time signals/ systems, uses; • Digital filters: FIR/IIR types, design, speech/ audio/radar signal processing uses, Microprocessors & microcontrollers, basics, interrupts, DMA, instruction sets, interfacing Controllers & uses; • Advance Embedded systems: Embedded Systems ARM Architecture, Embedded C
7	**Analog Communication Systems:** • Random Signals, noise, probability theory, information theory; • Analog versus digital communication and applications: systems – AM, FM, transmitters / receivers, theory/ practice/ standards, SNR comparison • Modulation Techniques : AM, FM & PM
8	**Digital Communication Systems:** • Digital communication basics; Sampling , quantizing, coding, PCM, DPCM, multiplexing- audio/ video, • Digital modulation Schemes: ASK, FSK, PSK QAM • Coherent – Non Coherent Detection • Error Probabilities in digital Modulation • Multiplexing and Multiple Access: Time Division Multiplexing (TDM), Frequency Division Multiplexing (FDM) Code Division Multiple Access (CDMA) Spread Spectrum Techniques. • Switching Techniques : Circuit Switching, Packet Switching & Message Switching
9	**Source Coding and Channel Capacity** • Entropy, mutual information • Shannon's theorem and channel capacity • Lossless source coding: Huffman coding, arithmetic coding
10	**Fading Channels and Diversity Techniques** Fading Channels : Rayleigh and Rician Fading BER over fading channels Diversity techniques : Time, Frequency and Space diversity MIMO systems and channel modeling.

11	**Advanced Digital Communication :** • Communication networks; Principles/ practices/ technologies/ uses/ OSI Model/ Security; • Basic Packet multiplexed streams/ scheduling; cellular networks, types, analysis, protocols (TCP/TCPIP); • Microwave & satellite communication; • Terrestrial / Space Type LOS systems; • Cognitive Radio and Spectrum sensing; • Cooperative Communication; • 5G and future communication systems. • block schematics link calculations, system design; • Communication satellite, orbits, characteristics, systems, uses; • Systems, uses, Fiber-optic communication systems, block schematics, link calculations, and system design.
12	**Control Systems :** • Basic Control System Components, Open loop and closed loop (Feedback) systems and stability analysis of these systems. • Tools and techniques for LTI control system analysis, Control system compensators, elements of lead and lag compensation, elements of Proportional Integral Derivative (PID) control. • Block Codes; Hamming code, Reed Solomon Codes • Convolutional Coding and Viterbi algorithm • Turbo codes, LDPC codes
13	**Communication Systems :** • Time and Frequency Domain Analysis • Power and energy spectral density of communication signals • Sampling and quantization • Line Coding, Pulse Shaping, Pulse and carrier modulation techniques • Transmitter and receiver design concepts
14	**Fiber Optics and Satellite :** • Introduction to Optical Fiber • Fiber Materials, Optical Sources and Photo Detectors, Fiber Fabrication • Optical couplers and WDM Concepts • System Design and Optical Fiber Applications • Satellite Subsystems and Transponder • Satellite Link Design • Earth Station Subsystems
15	**Wireless Broadband Communication:** • Wireless communication and their importance in age of high speed internet • Study of Wireless Network areas like WLAN and WWAN • Advanced 3G and 3G Wireless Communication • Communication System Security • RADAR Communication System • RF Devices and Applications • High Performance Network • Network Management
16	**Antennas:** • Types and Applications of Antennas.
17	**Line Communication** • EPABX (Analog & Digital) • IP EPABX Exchange (VoIP Based)
	Information Technology
18	**Basics of Information Technology:** • Introduction to IT: Definitions, applications and impact on Industries. • History and evolution of Computing • Computer Hardware and Software Overview • Operating Systems : Types, functions and features • Basic Networking Concepts • Internet Technologies: HEML,CSS and basic web development.
19	**Programming Fundamentals:** • Introduction to Programming Language (C. C++, Java or Python) • Variables, Data Types, Operators

	• Control Structures (if, else, loops) • Functions and Procedures • Object-Oriented Programming (OOP) Concepts • Data Structures: Arrays, Linked Lists, Stacks, Queues
20	**Database Management Systems (DBMS):** • Introduction to DBMS • Relational Database Concepts • SQL: Data Query Language (DQL), Data Manipulation Language (DML), Data Definition Language (DDL) • Normalization and ER Diagrams • Database Design and Architecture • Introduction to NoSQL databases
21	**Software Engineering:** • Software Development Life Cycle (SDLC) • Agile and Waterfall Models • Requirements Analysis and Specification • Software Design and Architecture • Testing and Debugging • Version Control Systems (Git)
22	**Data Structures and Algorithms:** • Complexity Analysis (Big O notation) • Sorting Algorithms: Bubble, Merge, Quick, etc. • Searching Algorithms: Binary Search, Linear Search • Trees, Graphs, and Hashing • Recursion • Dynamic Programming
23	**Computer Networks:** • OSI and TCP/IP • Layer functions: Application of Layer in the Internet • Application Layer – Host configuration, DHCP, Domain Name System (DNS) Multicast DNS • Network Protocols (HTTP, FTP, etc.) • IP Addressing and Subnetting • Wireless Networks and Mobile Computing • Network Security: Encryption, Firewalls, VPNs
24	**Information Security:** • Internet Security • Network layer security (AH, ESP, IPsec) • Transport layer security (SSL), Application layer security (Secure E mail- PGP, S/MIME) • Basics of Cyber security • Cryptography and Encryption Methods • Security Protocols and Standards • Risk Management and Incident Response • Ethical Hacking and Penetration Testing
25	**Web Technologies:** • Introduction to Web Development • Front-end Development (HTML, CSS, JavaScript) • Back-end Development (Node.js, PHP, Python) • Web Hosting and Domain Management • Web Security: SSL/TLS, HTTPS, Security Practices
26	**Cloud Computing and Virtualization:** • Introduction to Cloud Computing • Cloud Service Models (IaaS, PaaS, SaaS) • Virtualization Techniques • Cloud Deployment Models (Public, Private, Hybrid) • AWS, Azure, Google Cloud • Remote login TELNET and SSH, HTTPS, electronic mail
	Advanced and Emerging Technologies
27	**Emerging Technologies:** • Blockchain and Cryptocurrency

	• Internet of Things (IoT), IoT Protocols, • Advanced Topics and Industry-Oriented Learning Wireless Communication GSM, CDMA, LTE, 5G Technologies Nano technology • Big Data Analytics • Augmented Reality (AR) and Virtual Reality (VR) • Intrusion Detection System. • Multimedia Communication • Multimedia transport across IP networks and relevant protocols such as RSVP, RTP, RTCP, DVMRP, Signaling Protocols, Real Time Streaming Protocol (RTSP) • VoIP, IPTV • Information Representation – text, images, audio and video, Text and image compression, Audio and video compression, video • Compression standards: H.261, H.263, PI.323, MPEG 1, MPEG 2, other coding formats for text speech, image and video.
28	**Artificial Intelligence and Machine Learning (Basics):** • Introduction to AI and ML • Supervised, Unsupervised, and Reinforcement Learning • Popular Algorithms (Linear Regression, Decision Trees, etc.) • Applications of AI in Industry
29	**Modern digital communication systems in electronics and telecommunications engineering\ MIMO** • (Multiple Input Multiple Output): Uses multiple antennas at both the transmitter and receiver to improve communication performance. • Adaptive Modulation and Coding (AMC): Dynamically adjusts the modulation scheme and coding rate based on channel conditions to optimize performance. • Cognitive Radio: A smart communication system that dynamically adjusts its operating parameters based on the environment to optimize spectrum use.
30	**Advanced Communication** • VLSI Antenna and Wave Propagation Antenna Types, Radiation Patterns, Propagation Models VLSI Design • CMOS Technology • HDL • Logic Synthesis Computer Organization and Architecture Microprocessors, • Memory, • I/O Systems Power Electronics Converters, • Inverters, • Motor Drives, Real-Time Operating Systems (RTOS)
	Current Events:
31	• Current issues of national and international importance relating to social, economic, Political Cultural and Sports and telecommunication industrial developments. • Engineering Mathematics and Numerical Analysis • Standards and Quality practices in production, maintenance and services. • Information and Communication Technologies (ICT) based tools and their applications in Engineering such as Networking, e- governance and technology based education. • Knowledge of relevant topics as may be expected from an Engineering Graduate.
	Acts regarding Telecommunication:
32	1. Indian Wireless Telegraphy Act, 1933 2. Indian Telegraph Act, 1985 3. National Telecom Act, 2023

Note:- Wherever Acts/Rule/Regulations are mentioned in syllabus, latest amendments/versions to be considered .

दिनांक – ३० एप्रिल, २०२५

अवर सचिव
महाराष्ट्र लोकसेवा आयोग